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Skin Cancer Prevention (PDQ®)–Health Professional Version

Go to Patient Version

Who Is at Risk?

Individuals who have light-hair and -eye color, freckles, and who sunburn easily are particularly susceptible to developing skin cancer.[1] There are two primary types of skin cancer, keratinocyte carcinoma (including basal cell carcinoma and squamous cell carcinoma [SCC]) and melanoma. Observational and analytic epidemiological studies have consistently shown that increased cumulative sun exposure is a risk factor for keratinocyte carcinoma.[1,2] Melanoma risk correlates with common and atypical nevi.[3] Some studies suggest that there may be an interplay between genetic phenotype and sun exposure and that there may be two pathways to melanoma development.[4-7]

Organ transplant recipients taking immunosuppressive drugs are at an elevated risk of developing skin cancer, particularly SCC.[8,9] Arsenic exposure also increases the risk of keratinocytic cancers [10] and melanoma.[11]

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Overview

Note: The Overview section summarizes the published evidence on this topic. The rest of the summary describes the evidence in more detail.

Other PDQ summaries containing information related to skin cancer prevention include the following:

- Skin Cancer Screening
- Skin Cancer Treatment
- Genetics of Skin Cancer
- Levels of Evidence for Cancer Screening and Prevention Studies

Factors Associated With an Increased Risk of Keratinocyte Carcinoma (Basal Cell Carcinoma, Squamous Cell Carcinoma)

Fair skin

Based on solid evidence, individuals with fair skin types (light or pale skin, light-hair and -eye color, freckles, or those who burn easily) are associated with an increased risk of squamous cell carcinoma (SCC) and basal cell carcinoma (BCC).

Magnitude of Effect: Substantial, depending on the amount of exposure.

Study Design: Observational studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Sun and UV radiation exposure

Based on solid evidence, sun and UV radiation exposure are associated with an increased risk of SCC and BCC.

Magnitude of Effect: Substantial, depending on the amount of exposure.

Study Design: Observational studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Immunosuppression

Based on solid evidence, immunosuppression after organ transplant is associated with an increased risk of SCC and BCC.

Magnitude of Effect: Substantial, although not consistently quantitated.

Study Design: Observational studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Arsenic exposure

Based on fair evidence, arsenic exposure is associated with an increased risk of keratinocyte carcinoma.

Magnitude of Effect: Arsenic exposure is associated with keratinocyte carcinoma.

Study Design: One case-control study.

Internal Validity: Good.

Consistency: Fair.

External Validity: Fair.

Factors Associated With an Increased Risk of Melanoma

Sun and UV radiation exposure

Based on fair evidence, intermittent acute sun exposure leading to sunburn is associated with an increased risk of melanoma.

Magnitude of Effect: Unknown.

Study Design: Observational studies.

Internal Validity: Fair.

Consistency: Fair.

External Validity: Poor.

Arsenic exposure

Based on fair evidence, arsenic exposure is associated with an increased risk of melanoma.

Magnitude of Effect: Arsenic exposure is associated with double the incidence of melanoma.

Study Design: One case-control study.

Internal Validity: Good.

Consistency: Fair.

External Validity: Fair.

Interventions for Skin Cancer Prevention With Adequate Evidence

Treatment of sun-damaged skin to prevent skin cancer: Benefits

There is one well designed randomized controlled trial (RCT) that demonstrated the use of topical fluorouracil on sun-damaged skin prevents additional actinic keratoses and SCC requiring surgery.[1]

Magnitude of Effect: Moderate net benefit in preventing SCC requiring surgery.

Study Design: RCT.

Internal Validity: Good.

Consistency: N/A (single study).

External Validity: Fair.

Treatment of sun-damaged skin to prevent skin cancer: Harms

The primary side effect is local erythema, irritation, and crusting.

Interventions for Skin Cancer Prevention With Inadequate Evidence

Behavior counseling to change sun-protection practices: Benefits

Evidence from 21 RCTs demonstrated that behavior counseling for children and families and for adults improves sun protective behaviors. These trials showed an inconsistent effect on reducing sunburns and do not provide direct evidence on reduction of SCC, BCC, or melanoma.[2]

Magnitude of Benefit: Moderate net benefit for improving sun protective behaviors, but there is inadequate direct evidence to determine the impact on the development of skin cancer.

Study Design: Systematic review including 21 RCTs.

Internal Validity: Good.

Consistency: Good for behaviors. Poor for sunburns.

External Validity: Good.

Behavior counseling to change sun-protection practices: Harms

Avoiding sun exposure can result in harms, such as mood disorders, sleep disturbances, elevated blood pressure, and impaired vitamin D metabolism, which is associated with increased incidence of colon, ovary, and breast cancers, and multiple myeloma.[3]

Topical treatments to prevent skin cancer—sunscreen: Benefits

Sunscreen has been shown to prevent sunburns and actinic keratoses. RCTs showed inconsistent benefit in preventing SCC and showed no benefit in preventing melanoma.

Magnitude of Effect: Inadequate evidence to assess magnitude of effect for sunscreen.

Study Design: RCTs and observational cohort studies.

Internal Validity: Poor.

Consistency: Inconsistent.

External Validity: Poor.

Topical treatments to prevent skin cancer—sunscreen: Harms

Harms of sunscreen for the user are mild and mainly include skin allergic reactions. Because sunscreen use prevents sunburns, it may encourage more sun exposure to fair skinned people at risk of developing skin cancer.

Systemic treatments to prevent skin cancer (nonsteroidal anti-inflammatory drugs [NSAIDs], nicotinamide, isotretinoin, selenium, beta carotene, alpha-difluoromethylornithine [DFMO]): Benefits

There is no evidence showing that NSAIDs and nicotinamide prevent SCC. RCTs found no benefit in preventing SCC, BCC, or melanoma for topical or oral retinoids, selenium, and beta carotene. One RCT showed a slight reduction in BCC for DFMO, but no change in SCC or melanoma.

Magnitude of Effect: Inadequate evidence to assess magnitude of effect for topical retinoids, and nicotinamide. Harms likely outweigh potential benefits for NSAIDs, oral retinoids, beta carotene, and DFMO.

Study Design: RCTs and observational cohort studies.

Internal Validity: Poor.

Consistency: Inconsistent.

External Validity: Poor.

Systemic treatments to prevent skin cancer (NSAIDs, nicotinamide, isotretinoin, selenium, beta carotene, DFMO): Harms

NSAIDs are associated with adverse cardiovascular effects, gastrointestinal bleeding, and kidney damage. Oral retinoids are hepatotoxic and cause hypertriglyceridemia. In RCTs, beta carotene is associated with an increased risk of lung cancer incidence and mortality in smokers. Isotretinoin has dose-related skin toxicity. Patients discontinue DFMO at high rates because of hearing loss.

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Incidence and Mortality of Skin Cancer

There are two main types of skin cancer:

- Keratinocyte carcinoma.
    - Basal cell carcinoma (BCC).
    - Squamous cell carcinoma (SCC).
- Melanoma.

BCC and SCC are the most common forms of skin cancer but have substantially better prognoses than the less common, generally more aggressive, melanoma.

Keratinocyte carcinomas are the most commonly occurring cancer in the United States, but exact incidence figures are unavailable because cases are not required to be reported to cancer registries. Incidence rates appear to have been increasing for a number of years,[1] in part due to increased screening and biopsy of skin lesions. Based on an extrapolation of Medicare fee-for-service data to the U.S. population, about 3 million individuals were estimated to have been diagnosed with keratinocyte carcinomas in 2012,[1,2] exceeding all other cancer cases (approximately 2 million) estimated by the American Cancer Society in 2025.[1]

Melanoma cases are reported to U.S. cancer registries, so data are available. In 2025, an estimated 104,960 individuals in the United States will be diagnosed with melanoma and approximately 8,430 will die of the disease.[1] While only 2% of skin cancers are melanomas, melanoma causes more than 80% of deaths from skin cancer.[3]

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Accuracy of Making a Clinical Diagnosis of Melanoma

Observer variability among physicians has been noted in the evaluation of skin lesions and subsequent biopsy specimens. A systematic review of 32 studies that compared the accuracy of dermatologists and primary care physicians in making a clinical diagnosis of melanoma concluded that there was no statistically significant difference in accuracy. However, the results were inconclusive, owing to small sample sizes and study design weaknesses.[1] Subsequent studies have noted a higher accuracy for dermatologists in the diagnosis of melanocytic lesions,[2,3] yet there is a shortage of dermatologists to meet the demands of population-level screening.

A study of 187 pathologists in the United States found that cases of moderately dysplastic nevi to early-stage invasive melanoma had less than 50% agreement with a reference diagnosis defined by consensus of experienced pathologists.[4] At a U.S. population level, it is estimated that 82.8% (95% confidence interval, 81.0%–84.5%) of melanocytic skin biopsy diagnoses would be verified if they were reviewed by a consensus reference panel of experienced pathologists.[4] In addition, differentiating between benign and malignant melanocytic tumors during histological examinations of biopsy specimens has been shown to be inconsistent, even in the hands of experienced dermatopathologists.[5,6] This variability in the diagnosis of melanocytic lesions undermines the results of studies that examine screening effectiveness and also may undermine the effectiveness of any screening intervention. Furthermore, this finding suggests that requesting a second opinion regarding the pathology of biopsy specimens may be important.[5-7] A standard approach to the classification of melanocytic skin lesions by pathologists may also reduce confusion and improve communication between clinicians.[4,6,8,9]

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Risk Factors for Skin Cancer

Epidemiological evidence suggests that exposure to UV radiation and the sensitivity of an individual's skin to UV radiation are the main risk factors for skin cancer, although the type of exposure (high-intensity and short-duration vs. chronic exposure) and the pattern of exposure (continuous vs. intermittent) may differ among the two main skin cancer types.[1-3]

The immune system plays a role in the pathogenesis of skin cancer: organ transplant recipients taking immunosuppressive drugs are at an elevated risk of skin cancer, both squamous cell carcinoma (SCC) and melanoma.[4] Arsenic exposure also increases the risk of cutaneous SCC.[4]

The visible evidence of susceptibility to skin cancer (skin type and precancerous lesions), presence of sun-induced skin damage (sunburn and solar keratoses), and increased number of nevi and atypical nevi are associated with an increased risk of melanoma.[5,6]

Factors Associated With Increased Risk of Keratinocyte Carcinoma

UV radiation exposure

Most evidence about UV radiation exposure and the prevention of skin cancer comes from observational and analytic epidemiological studies. Such studies have consistently shown that increased cumulative sun exposure is a risk factor for keratinocyte carcinomas.[2,3] Individuals whose skin tans poorly or burns easily after sun exposure are particularly susceptible.[2]

Actinic keratoses

It is generally felt that one-half or more of SCCs arise from actinic keratoses. However, nearly one-half of SCCs occur in clinically normal skin.[7] A longitudinal study has shown that the progression rate from actinic keratoses to SCC is about 0.075% to 0.096% per year, or less than 1 case in 1,000 per year.[7] Moreover, in a population-based longitudinal study, there was an approximately 26% spontaneous regression rate of actinic keratoses within 1 year of a screening examination.[8]

Factors Associated With an Increased Risk of Melanoma

UV radiation exposure

The relationship between UV radiation exposure and cutaneous melanoma is less clear than the relationship between UV exposure and keratinocyte carcinoma. In the case of melanoma, it seems that intermittent acute sun exposure leading to sunburn is more important than cumulative sun exposure;[9] such exposures during childhood or adolescence may be particularly important.[1]

Multiple case control studies have also documented the association between sun exposure and melanoma. Total sun exposure in childhood is associated with an increased risk of melanoma (odds ratio, 1.81–4.4) as is recreational sun exposure during childhood and adulthood, while occupational sun exposure may be associated with a decreased risk of melanoma.[10,11] Fair skin that sunburns easily has a twofold risk of melanoma compared with skin phenotypes that never burn. Natural red and blond hair and natural blond hair also confers a twofold to fourfold increased risk of melanoma.[12]

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Interventions for Skin Cancer Prevention With Adequate Evidence of Benefit

Treatment of Sun-Damaged Skin to Prevent Skin Cancer

Topical fluorouracil

Daily application of topical fluorouracil for up to 4 weeks onto actinic keratosis has been shown to reduce the development of new actinic keratoses.[1,2] A randomized controlled trial included 932 veterans with sun-damaged skin (two or more keratinocyte carcinomas in the 5 years before enrollment). Participants were randomly assigned to a 2- to 4-week single course of 5% topical fluorouracil or vehicle control cream. The fluorouracil group had fewer actinic keratosis cases when compared with the control group at 6 months (3.0 vs. 8.1; $P < .001$) and for the overall study duration ($P < .001$). Topical fluorouracil also reduced the risk of squamous cell carcinoma (SCC) requiring surgery at those sites for 1 year, but no effect was seen on basal cell carcinoma (BCC) in year 1 or on SCC or BCC over 4 years. Erythema, crusting, and irritation were reported by 82% of participants, with 40% reporting symptoms as severe. However, at study completion, 87% reported a willingness to repeat the treatment if needed.[1]

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Interventions for Skin Cancer Prevention With Inadequate Evidence of Benefit

Behavioral Interventions to Change Sun-Protective Practices

The U.S. Preventive Services Task Force (USPSTF) commissioned a systematic review of primary care behavioral counseling interventions for skin cancer prevention.[1] The review identified 21 trials on promoting protective behaviors in 27 publications with 20,561 participants. Protective behaviors included use of protective clothing to limit UV radiation exposure, sun-avoidance behaviors, and use of sunscreen. Interventions included physician counseling, tailored mailings and texts, educational presentations, and interactive web programs involving patients and families. Five of six trials in children found that interventions reduced parent-reported composite sun protection scores at 3 months to 3 years.[2-6] Six of twelve trials in adults also showed that interventions resulted in a reduced patient-reported composite sun protection score, with the greatest change being increased use of sunscreen.[7-13]

The trials did not show a consistent change in sunburns for children or adults. In the three trials of children that assessed changes in sunburn frequency (n = 2,508),[14-16] only one trial showed a reduction in nonsevere burns, but no change in severe burns.[15] In the six trials of adults that assessed changes in sunburn frequency (n = 3,959),[17-22] only one trial showed a slight reduction in red or painful burns at 3 months.[17] There were no changes reported in any trial showing reductions in skin cancer (keratinocyte carcinoma or melanoma) or skin cancer precursors (nevi or actinic keratosis).

While direct evidence is lacking, the USPSTF linked the evidence demonstrating that behavioral counseling interventions promote sun protective practices with the epidemiological data on UV exposure and skin cancer prevalence. This led to a recommendation for counseling children, adolescents, and young adults aged 6 months to 24 years and adults older than 24 years with fair skin on protective practices to reduce skin cancer.[23]

Topical Treatment to Prevent Skin Cancer—Sunscreen

Sunscreen use has been shown to decrease the rate of developing new actinic keratoses [24] and to increase the remission rate of existing lesions.[25] Another trial found no difference in keratinocyte cancers in daily versus discretionary sunscreen users.[26] An 8-year observational posttrial follow-up showed reductions in both squamous cancers [27] and melanomas [28] associated with sunscreen use, but the confidence intervals (CIs) were very wide, and the participation outside the initial trial introduced uncertainty.

A meta-analysis of 18 studies that explored the association between melanoma risk and previous sunscreen use illustrated widely differing study qualities and suggested little or no association.[29] A systematic review of the association between sunscreen use and the development of melanocytic nevi in children reported similar issues with study quality and heterogeneity, hindering conclusive assessments. However, of the 15 studies that met inclusion criteria, 12 found either an increased incidence or no association.[30]

Systemic Medications to Prevent Skin Cancer

Nonsteroidal anti-inflammatory drugs (NSAIDS)

A randomized controlled trial (RCT) included 240 people at high risk of skin cancer (each with 10–40 actinic keratoses and a history of previous skin cancer) who were given celecoxib 200 mg twice daily or a placebo for 9 months. The trial found no difference in the incidence of actinic keratosis, but a post hoc analysis revealed a statistically significant difference in the mean number of keratinocyte carcinomas per patient (rate ratio, 0.43; 95% CI, 0.24–0.75; absolute difference, 0.2 lesions per patient).[31] A meta-analysis of nine studies (five case-control, three cohort, and one intervention) reported a small reduction in squamous cell carcinoma (SCC) risk associated with the use of nonaspirin NSAIDs (relative risk [RR], 0.85; 95% CI, 0.78–0.94), with the effect seen particularly in those with previous actinic skin tumors.[32]

NSAIDs are associated with known adverse cardiovascular effects, gastrointestinal bleeding, and kidney damage.[33]

Nicotinamide (vitamin B3)

The effect of nicotinamide on the development of new actinic keratosis lesions has been studied with inadequate evidence for efficacy, even in higher-risk populations. Studies include a clinical trial of patients with four or fewer actinic keratosis lesions at baseline (Oral Nicotinamide to Reduce Actinic Cancer [ONTRAC]) [34] and a trial of immunosuppressed organ-transplant recipients (Oral Nicotinamide to Reduce Actinic Cancer after Transplant [ONTRANS]).[35] The ONTRAC trial showed a lower rate of new lesions while individuals received treatment, but not during the 6-month postintervention follow-up period.[36] The ONTRANS trial was impacted by slow recruitment and was stopped early but showed no efficacy in the limited sample size.

Isotretinoin and related systemic retinoids such as acitretin

Retinoids are vitamin A derivatives that are available in topical and oral preparations. Oral retinoids have been studied in high-risk populations, such as those with a history of multiple nonmelanoma skin cancers, genetic disorders such as xeroderma pigmentosum, transplant recipients, and those exposed to high cumulative levels of psoralen plus UV A (PUVA) therapy.[37-43] However, side effects of oral retinoids, including hypertriglyceridemia and hepatic toxicity, are significant.

Topical tretinoin 0.1% cream was compared with a control for 1.5 to 5.5 years in an RCT. No difference was found in the proportions of patients who developed SCC or basal cell carcinoma (BCC) or actinic keratosis.[44]

Selenium

A multicenter, double-blind, randomized, placebo-controlled trial included 1,312 patients with a history of BCC or SCC and a mean follow-up of 6.4 years. The study showed that 200 µg of selenium (in brewer’s yeast tablets) did not have a statistically significant effect on the primary end point of BCC development, but selenium instead increased the risk of SCC and total keratinocyte carcinomas (unadjusted RR, 1.27; 95% CI, 1.11–1.45).[45,46]

Beta carotene

In the Physicians’ Health Study, 21,884 male physicians with no reported history of BCC or SCC were randomly assigned to take 50 mg doses of daily oral beta carotene versus placebo in a 2 × 2 factorial trial of beta carotene and aspirin.[47] After 12 years, there was no difference in incidence of either BCC or SCC between the beta carotene and placebo groups. Similar findings were noted in 10 years and 14 years of follow-up among the participants in the Nurses' Health Study and the Health Professionals Follow-up Study.[48] Reanalysis of data from these two cohorts after an additional 16 years of follow-up noted higher intake of some carotenoids was associated with a modest reduction in SCC risk.[49] Data on the use of sun protection behaviors were not available, and as participants with higher intake tended to have higher levels of physical activity, lower smoking, and alcohol consumption, it is possible that there was a confounding effect of sun protection behaviors. RCTs of long-term treatment with beta carotene in individuals previously treated for keratinocyte carcinoma also showed no benefit in preventing the occurrence of new keratinocyte carcinomas.[26,50]

Several RCTs show that beta carotene supplementation can increase cardiovascular disease mortality and increase the risk of lung cancer.[51,52]

Alpha-difluoromethylornithine (DFMO)

An RCT of oral DFMO (500 mg/m^2/day) versus placebo for up to 5 years (n = 250 participants) showed no difference in the number of new keratinocyte carcinomas.[53] A subset analysis showed a difference in BCC events favoring the DFMO group (0.28 vs. 0.40 per year; P = .03) but no difference in SCC rates. However, the DFMO group experienced greater hearing loss than the placebo group (4 dB vs. 2 dB, P = .003), resulting in a higher study drug discontinuation rate (10.8% vs. 4.5%).

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Latest Updates to This Summary (04/08/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Incidence and Mortality of Skin Cancer

Added American Cancer Society as reference 1.

Updated statistics with estimated new cases and deaths of melanoma for 2025.

This summary is written and maintained by the PDQ Screening and Prevention Editorial Board, which is editorially independent of NCI. The summary reflects an independent review of the literature and does not represent a policy statement of NCI or NIH. More information about summary policies and the role of the PDQ Editorial Boards in maintaining the PDQ summaries can be found on the About This PDQ Summary and PDQ® Cancer Information for Health Professionals pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about skin cancer prevention. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

This summary is reviewed regularly and updated as necessary by the PDQ Screening and Prevention Editorial Board, which is editorially independent of the National Cancer Institute (NCI). The summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

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